

MARYAM RAEES AHMED

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PROFESSIONAL SUMMARY

Final-year BS Computer Science student (CGPA: 3.4/4.0) with hands-on experience in Machine Learning, Data Science, and Data Engineering basics. Built real-world projects involving large datasets, predictive modeling, deep learning, and recommendation systems. Strong in Python, SQL, and PyTorch, with practical experience in data preprocessing, model training, and evaluation. Interested in applying AI/ML skills to solve real industry problems in data-driven environments.

EDUCATION

Dawood University of Engineering & Technology

BS Computer Science | CGPA: 3.4 / 4.0

2026

Karachi, Pakistan

TECHNICAL SKILLS

Programming Languages: Python, SQL

Data Science & Machine Learning: PyTorch, KNN, Time-Series Forecasting, Predictive Modelling, Recommendation Systems

Databases & Data Management: MySQL, Relational Database Design, Data Modelling

CERTIFICATIONS

Certified Data Scientist | PITP – NED University

Python, Data Analysis, Machine Learning (Regression, Classification, Clustering...), Scikit-learn, Pandas

2024

PROJECTS

ZARKHAIZ – Intelligent Smart Irrigation Framework (Final Year Project)

Python · PyTorch · LSTM · GNN

Oct 2025 – June 2026

Final Year Project

Built a large-scale time-series dataset (45,984 hourly records, 16M+ rows) using Open-Meteo API, engineered for multivariate soil moisture forecasting.

Developed and trained a deep learning forecasting model (LSTM + Graph Attention Networks + GRU, Seq2Seq architecture) with a physics-aware loss function for improved predictive reliability.

Conducted model benchmarking and ablation studies (LSTM vs JK-GCN vs JK-GAT) achieving MAE: 0.3360%, R^2 : 0.99+, and +25.88% Forecast Skill Score improvement over baseline methods.

Alternox – Medicine Alternative Recommender

Python · Flask · Cosine Similarity

Dec 2025 – Feb 2026

Research Project

Built an AI/ML-based recommendation system (Python) that identifies affordable substitute medicines using pharmacological similarity from a structured Kaggle dataset (ingredients, dosage, price, manufacturer).

Developed a hybrid similarity engine using KNN and Cosine Similarity to rank alternative medicines based on chemical composition and cost efficiency.

Implemented a Flask-based web application with search, authentication, and real-time recommendation output, delivering a working proof-of-concept for cost-effective medicine substitution..

UniSwap – University Item Exchange Platform

MySQL · ERD · SQL

May 2025 – Jul 2025

Designed and implemented a fully normalized relational database (MySQL) with structured entities (Users, Items, Categories), ensuring data integrity, scalability, and minimal redundancy through 3NF principles.

Created a complete ERD (dbdiagram.io) and engineered primary/foreign key relationships to model real-world system logic and enforce strict referential integrity across all tables.

Optimized schema for efficient querying and structured data extraction, enabling reliable filtering, aggregation, and analytics-ready data organization.